# Introduction to Internet of Things

## Assignment-Week 1

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

IoT stands for _____.

a. Internet of Tasks
b. Internet of Tuples
c. Internet of Things
d. None of these

**Correct Answer: c. Internet of Things**

**Detailed Solution:** The full form of IoT is "Internet of Things"

See lecture 1 (Introduction to IoT – Part - I) @ 1:30

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**QUESTION 2:**

Which of the following technologies have unified and has resulted in the evolution of IoT?

a. High-power embedded systems
b. Super Computing
c. Engine Technology
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution**: Unification of technologies which has resulted in the advancement of IoT are –

**a.** Low-power embedded systems

**b.** Cloud Computing

**c.** Big Data

**d.** Machine Learning

**e.** Networking

See lecture 1 (Introduction to IoT – Part - I) @ 5:54

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**QUESTION 3:**

Which of the following are the enables of IoT?

a. RFID
b. Nanotechnology
c. Sensors
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** The enables of IoT are –

a. RFID

b. Nanotechnology

c. Sensors

d. Smart Networks

See lecture 2 (Introduction to IoT – Part - I) @ 12:50

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**QUESTION 4:**

Which of the following is NOT a function of an IoT Gateway?

a. Connects the IoT LAN to a WAN

**b.** Can implement several LAN and WAN

**c.** Both (a) and (b)

**d.** Neither (a) Nor (b)

**Correct Answer: d. Neither (a) nor (b)**

**Detailed Solution:** The functionalities of an IoT Gateway are –

a. Connects the IoT LAN to a WAN

b. Can implement several LAN and WAN

c. Forwards packets between LAN and WAN on the IP layer

See lecture 2 (Introduction to IoT – Part - II) @ 3:09

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**QUESTION 5:**

State whether the following statement is True or False.

Statement: "The ITU vision is approaching reality as the present-day networked devices have outnumbered humans on earth."

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** The ITU vision is approaching reality as the present-day networked

devices have outnumbered humans on earth.

See lecture 2 (Introduction to IoT – Part - II) @ 02:35

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**QUESTION 6:**

State True or False.

Statement: "In Multi-homing, a node/network is connected to multiple networks for improved reliability.

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** In multi-homing, a node/network is connected to multiple networks for improved reliability.

See lecture 2 (Introduction to IoT – Part - II) @ 15:22

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**QUESTION 7:**

Which of the following are the features of an IoT Gateway?

a. A router connecting IoT LAN and WAN to Internet
b. Can implement several LAN and WAN
c. Forwards packets between LAN and WAN
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** The following are the features of an IoT gateway –

- A router connecting IoT LAN and WAN to the Internet
- Can implement several LAN and WAN

- Forwards packets between LAN and WAN

See lecture 2 (Introduction to IoT – Part - II) @ 04:00

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**QUESTION 8:**

State True or False.

Statement: "IoT Proxy performs active application layer functions between IoT nodes and other entities."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** IoT Proxy performs active application layer functions between IoT nodes and other entities.

See lecture 2 (Introduction to IoT – Part - II) @ 04:00

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**QUESTION 9:**

The Gateway has a _____ address prefix, which can be used to identify them globally.

a. Non-unique
b. Unique
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: b. Unique**

**Detailed Solution:** The Gateway has a unique address prefix, which can be used to identify

them globally.

See lecture 2 (Introduction to IoT – Part - II) @ 07:00

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**QUESTION 10:**

The IPv4 notation uses _____ number of bits to represent an address.

e. 33
f. 32
g. Both (a) and (b)
h. Neither (a) nor (b)

**Correct Answer: b. 32**

**Detailed Solution:** The IPv4 notation uses 32 bits to represent an address.

See lecture 2 (Introduction to IoT – Part - II) @ 16:33

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**QUESTION 11:**

The "Destination Address" field in the IPv4 header is of _____ bits in length.

a. 32
b. 64
c. 128
d. None of these

**Correct Answer: a. 32**

**Detailed Solution:** The IPv4 uses 32 bits to represent the "Destination Address".

See lecture 2 (Introduction to IoT – Part - II) @ 17:53

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**QUESTION 12:**

We can sense temperature using _____?

a. An actuator
b. A Transducer
c. A processor
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** We can sense temperature using a Sensor.

See lecture 3 (Sensing) @ 01:00

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**QUESTION 13:**

State True or False.

Statement: "A sensor is not only sensitive to the measured property, but it also influences the measured property."

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** A sensor is only sensitive to the measured property. It does not influence the measured property.

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See lecture 3 (Sensing) @ 12:11

**QUESTION 14:**

Based on the output, sensors are classified as _____.

a. Scalar and Vector
b. Analog and Digital
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: b. Analog and Digital**

**Detailed Solution:** Based on the output, sensors are classified as Digital and Analog.

See lecture 3 (Sensing) @ 13:05

**QUESTION 15:**

What is the full form of SMP in the context of IoT actuators?

a. Shape Master Polymers
b. Switch Mode Power
c. Sensor Making Polymer
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** The full form of SMP is Shape Memory Polymers.

See Page number – 111, Chapter - 5, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

**********END**********

# Introduction to Internet of Things

## Assignment-Week 2

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

Based on functionality, MQTT is a _____ protocol.

a. Transport

b. Data
c. Semantic
d. None of these

**Correct Answer: b. Data**

**Detailed Solution:** MQTT is a Data Protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 01:30

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**QUESTION 2:**

MQTT is a _____-based lightweight protocol for use in conjunction with TCP/IP protocol.

a. Client-Server
b. Publish-Subscribe
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: b. Publish-Subscribe**

**Detailed Solution:** MQTT, is a publish-subscribe-based lightweight protocol for use in conjunction with TCP/IP protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 02:12

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**QUESTION 3:**

A _____ controls the publish-subscribe messaging pattern in MQTT.

a. Publishers
b. Subscribers
c. Broker
d. None of these

**Correct Answer: c. Broker**

**Detailed Solution:** A Broker controls the publish-subscribe messaging pattern in MQTT.

See lecture 6 (Basics of IoT Networking – Part II) @ 03:31

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**QUESTION 4:**

State True or False.

**Statement:** "In MQTT, the Publishers are Lightweight Sensors."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** In MQTT, the Publishers are lightweight sensors.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:49

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**QUESTION 5:**

Which of the following is MQTT designed for?

a. Remote connections
b. Limited bandwidth
c. Small code footprint
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** MQTT is designed for –

a. Remote connections
b. Limited bandwidth
c. Small code footprint

See lecture 6 (Basics of IoT Networking – Part II) @ 03:23

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**QUESTION 6:**

A _____ to which a client is subscribed is updated in the form of messages and distributed by the _____.

a. Topic, Message Broker
b. Message Broker, Topic
c. Message Broker, Message Broker
d. Topic, Topic

**Correct Answer: a. Topic, Message Broker**

**Detailed Solution:** A topic to which a client is subscribed is updated in the form of messages and distributed by the message broker.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:00

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**QUESTION 7:**

There are _____ number of methods in MQTT protocol.

a. 5
b. 4
c. 6
d. None of these

**Correct Answer: a. 5**

**Detailed Solution:** There are 5 number of methods in MQTT protocol.

See lecture 6 (Basics of IoT Networking – Part II) @ 05:49

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**QUESTION 8:**

State True or False.

**Statement:** "Publish/Subscribe in MQTT is event-driven and enables messages to be pushed to clients."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** Publish/Subscribe in MQTT is event-driven and enables messages to be pushed to clients.

See lecture 6 (Basics of IoT Networking – Part II) @ 08:32

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**QUESTION 9:**

SMQTT stands for?

a. Secure MQTT
b. Safe MQTT
c. Safety in MQTT
d. Stored MQTT

**Correct Answer: a. Secure MQTT**

**Detailed Solution:** SMQTT stands for Secure MQTT.

See lecture 6 (Basics of IoT Networking – Part II) @ 14:10

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**QUESTION 10:**

Similar to HTTP, CoAP utilizes which of the following for operation?

a. GET
b. PUT
c. PUSH
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE message requests to retrieve, create, update, and delete, respectively.

See lecture 7 (Basics of IoT Networking – Part III) @ 08:24

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**QUESTION 11:**

State True or False.

**Statement:** "The CoAP protocol is a Client-Server interaction-based protocol that is asynchronous over a datagram-oriented transport protocol such as UDP."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** The CoAP protocol is a Client-Server interaction-based protocol that is asynchronous over a datagram-oriented transport protocol such as UDP.

See lecture 7 (Basics of IoT Networking – Part III) @ 00:30

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**QUESTION 12:**

What is the full form of M2M in the context of IoT protocol?

a. Message to Message
b. Man to Message
c. Message to Man
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** M2M stands for Machine to Machine.

See lecture 7 (Basics of IoT Networking – Part IV) @ 0:30

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**QUESTION 13:**

What is the basic unit of data in the AMQP protocol?

a. Chunk
b. Byte
c. Slit
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** In AMQP, the basic unit of data is called a frame.

See lecture 8 (Basics of IoT Networking – Part IV) @ 0:41

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**QUESTION 14:**

State True or False.

**Statement:** "The OSI model has 9 layers."

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** The OSI model is a conceptual framework that divides any networked communication system into seven layers.

See Page number – 10, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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**QUESTION 15:**

The "Destination Address" in the IPv4 packet represents which of the following?

a. The destination node address of the packet
b. The intermediate hop in the network
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: a. The destination node address of the packet**

**Detailed Solution:** The "Destination Address" in the IPv4 packet represents the address of the destination node in the network.

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**********END**********

# Introduction to Internet of Things

## Assignment-Week 3

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

State True or False.

**Statement:** "Wireless HART is the latest release of Highway Addressable Remote Transducer protocol."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** WirelessHART is the latest release of Highway Addressable Remote Transducer protocol.

See lecture 11 (Connectivity Technologies-III) @ 01:42

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**QUESTION 2:**

State True or False.

**Statement**: "Wired HART does not have a network layer."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** Wired HART does not have a network layer.

See lecture 11 (Connectivity Technologies-III) @ 04:25

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**QUESTION 3:**

Wireless HART physical layer is derived from _____ protocol?

a. 802.15.4
b. 802.15.2
c. 802.15.3
d. 802.15.6

**Correct Answer: a. 802.15.4**

**Detailed Solution:** WirelessHART physical layer is derived from 802.15.4 protocol.

See lecture 11 (Connectivity Technologies-III) @ 06:00

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**QUESTION 4:**

Wireless HART operates only in _____ GHz ISM band.

a. 3.7
b. 4.8
c. 4.8
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** HART operates only in the 2.4 GHz ISM band.

See lecture 11 (Connectivity Technologies-III) @ 06:00

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**QUESTION 5:**

Wireless HART employs and exploits _____ channels of the ISM band to increase reliability.

a. 13
b. 4
c. 2
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** WirelessHART employs and exploits 15 channels of the ISM band to increase reliability.

See lecture 11 (Connectivity Technologies-III) @ 06:00

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**QUESTION 6:**

State True or False.

**Statement:** "Channel blacklisting identifies channels consistently affected by interference and adds them from use again."

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** Channel blacklisting identifies channels consistently affected by interference and removes them from use.

See lecture 11 (Connectivity Technologies-III) @ 07:05

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**QUESTION 7:**

The Wireless HART incorporates _____ and _____ to increase the reliability and security of the network.

a. Channel hopping and Channel Blacklisting
b. Channel Aggregating and Frequency Aggregating
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: a. Channel hopping and Channel Blacklisting**

**Detailed Solution:** The WirelessHART incorporates Channel hopping and channel blacklisting to increase the reliability and security of the network.

See lecture 11 (Connectivity Technologies-III) @ 07:30

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**QUESTION 8:**

State whether the following statement is True or False

**Statement:** "Wireless HART protocol relies on Mesh Networking for its communication, and each device is primed to forward packets from every other device.".

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** WirelessHART protocol relies on Mesh Networking for its communication, and each device is primed to forward packets from every other device.

See lecture 11 (Connectivity Technologies-III) @ 09:00

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**QUESTION 9:**

State True or False.

**Statement:** "ZigBee protocol does not feature channel hopping at all."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** ZigBee protocol does not feature channel hopping at all.

See lecture 11 (Connectivity Technologies-III) @ 14:55

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**QUESTION 10:**

NFC is designed for use by devices within _____ to each other.

a. Close proximity
b. Anywhere on the globe
c. A small building
d. None of these

**Correct Answer: a. Close proximity**

**Detailed Solution:** NFC is designed for use by devices within close proximity to each other.

See lecture 11 (Connectivity Technologies-III) @ 17:47

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**QUESTION 11:**

NFC are of _____ types?

a. 2
b. 3
c. 1
d. None of these

**Correct Answer: a. 2**

**Detailed Solution:** NFC are of 2 types.

See lecture 11 (Connectivity Technologies-III) @ 18:47

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**QUESTION 12:**

State True or False.

**Statement**: "Smartphones are an example of active NFC devices."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** Smartphones are an example of active NFC devices.

See lecture 11 (Connectivity Technologies-III) @ 18:47

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**QUESTION 13:**

Bluetooth technology is based on _____.

a. HART
b. ZigBee
c. Ad-hoc piconets
d. None of these

**Correct Answer: c. Ad-hoc piconets**

**Detailed Solution:** Bluetooth technology is based on Ad-hoc piconets.

See lecture 12 (Connectivity Technologies-IV) @ 02:30

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**QUESTION 14:**

State whether the following statement is true or false.

**Statement:** The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

See Page number – 157, Chapter - 7, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

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**QUESTION 15:**

Zigbee commonly uses _____data rate.

a. 250 Mbps
b. 260 kbps
c. 260 Mbps
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** Zigbee commonly uses a 250 kbps data rate.

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**********END**********

# Introduction to Internet of Things

## Assignment-Week 4

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

State whether the following statement is true or false.

**Statement:** Coverage in WSN is defined as the entire area is covered satisfactorily.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** Coverage in WSN is defined as the area-of-interest is covered satisfactorily.

See lecture 17 (Sensor Networks-IV) @ 01:44

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**QUESTION 2:**

State True or False.

Statement: "In Wireless Sensor Network coverage, we study how to deploy or activate sensors to cover the monitoring area."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** In Wireless Sensor Network coverage, we study how to deploy or activate sensors to cover the monitoring area.

See lecture 17 (Sensor Networks-IV) @ 02:00

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**QUESTION 3:**

Water reservoir overflow monitoring in WSN is an example of which type of reporting?

a. Event driven
b. On-demand
c. Scattered
d. None of these

**Correct Answer: a. Event driven**

**Detailed Solution:** Water reservoir overflow monitoring in WSN is an example of event driven reporting.

See lecture 17 (Sensor Networks-IV) @ 08:13

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**QUESTION 4:**

State whether the following statement is true or false.

**Statement:** Objective of coverage in WSN is to use minimum number of sensors and minimize network lifetime.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution:** The objective of coverage in WSN is to use minimum number of sensors and maximize the network lifetime.

See lecture 17 (Sensor Networks-IV) @ 08:47

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**QUESTION 5:**

State whether the following statement is true or false.

**Statement:** "Sensor nodes in WSN can be of two types – 1. Static sensor nodes, and 2. Mobile sensor nodes."

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** Sensor nodes in WSN can be of only two types – 1. Static sensor nodes, and 2. Mobile sensor nodes.

See lecture 17 (Sensor Networks-IV) @ 02:00

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**QUESTION 6:**

An inventory control system is an example of _____?

a. Event-driven reporting
b. On-demand reporting
c. No sensing and reporting involvement
d. None of these

**Correct Answer: b. On-demand reporting**

**Detailed Solution:** An inventory control system is an example of On-demand reporting.

See lecture 17 (Sensor Networks-V) @ 08:40

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**QUESTION 7:**

Most problems in static WSN can be classified as -

a. Area coverage
b. Barrier coverage
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: c. Both (a) and (b)**

**Detailed Solution:** Most problems in static WSN can be classified as -

a. **Area coverage**
b. **Point coverage**
c. **Barrier coverage**

See lecture 17 (Sensor Networks-V) @ 11:30

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**QUESTION 8:**

What are the disadvantages of Stationary WSN?

a. Node failure may result in the partition of networks
b. Topology cannot be changed automatically
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: c. Both (a) and (b)**

**Detailed Solution:** The disadvantages of Stationary WSN are -

a. Node failure may result in the partition of networks
b. Topology cannot be changed automatically

See lecture 18 (Sensor Networks-V) @ 00:52

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**QUESTION 9:**

State True or False.

**Statement**: "MANET-infrastructure less network of mobile devices connected wirelessly in WSN follow the self-HOP properties."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** MANET-infrastructure-less network of mobile devices connected wirelessly in WSN follow the self-HOP properties.

See lecture 18 (Sensor Networks-V) @ 04:44

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**QUESTION 10:**

In Mobile WSN, the Data Mules

a. Collect the data from sensor nodes
b. Goes to the sink and delivers the collected data
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: c. Both (a) and (b)**

**Detailed Solution:** In Mobile WSN, the Data Mules

a. Collect the data from sensor nodes
b. Goes to the sink and delivers the collected data

See lecture 18 (Sensor Networks-V) @ 07:15

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**QUESTION 11:**

_____ senses different under the sea or water level parameters.

a. UAV
b. AUV
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: b. Autonomous Underwater Vehicle (AUV)**

**Detailed Solution: AUV** senses different under the sea or water level parameters.

See lecture 18 (Sensor Networks-V) @ 08:35

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**QUESTION 12:**

Humans carry their devices and move around. Sensors embedded within the devices record readings. Sensory readings are then transmitted for processing.

This paradigm of sensing is known as –

a. Machine Centric Sensing
b. Device Centric Sensing
c. Human Centric Sensing
d. None of these

**Correct Answer: c. Human Centric Sensing**

**Detailed Solution:** Humans carry their devices and move around. Sensors embedded within the devices record readings. Sensory readings are then transmitted for processing.
This paradigm of sensing is known as Human Centric Sensing.

See lecture 18 (Sensor Networks-V) @ 11:37

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**QUESTION 13:**

State True or False.

**Statement:** "Energy of Devices and Participant selection are two major problems in Human Centric Sensing."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** The following are the challenges in human-centric sensing -

a. Energy of devices
b. Participant selection
c. Privacy of users

See lecture 18 (Sensor Networks-V) @ 12:45

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**QUESTION 14:**

Which of the following network topologies is used in UAV networks?

a. Mesh
b. Star
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: a. Both (a) and (b)**

**Detailed Solution:** UAV networks use the mesh and star network topologies.

See lecture 19 (UAV Networks) @ 02:43

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**QUESTION 15:**

The _____ provides integrated services based on device collected data-sets.

a. M2M Application Platform
b. M2M Service Platform
c. M2M Hardware Platform
d. None of these

**Correct Answer: a. M2M Application Platform**

**Detailed Solution:** The M2M Application Platform provides integrated services based on device collected data-sets.

See lecture 20 (Machine to Machine Communication) @ 18:00

**********END**********

# Introduction to Internet of Things

## Assignment-Week 5

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

When different processing logic are applied to the same IoT networked devices or applications, it gives rise to?

a. Large scale of co-operation

b. Semantic conflicts

c. Both (a) and (b)

d. Neither (a) nor (b)

**Correct Answer: b. Semantic conflicts**

**Detailed Solution: When different processing logic are applied to the same IoT networked devices or applications, it gives rise to Semantic conflicts.**

See lecture 21 (Interoperability in Internet of Things) @ 03:41.

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**QUESTION 2:**

State True or False.

**Statement:** "Interoperability is a characteristic of a product or system."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** Interoperability is a characteristic of a product or system.

Refer Lecture 21@5:51

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**QUESTION 3:**

Interoperability is important in the context of IoT so that -

a. Physical objects remain isolated
b. Networks remain isolated
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: d. Neither (a) nor (b)**

**Detailed Solution:** Interoperability is important in the context of IoT so that –

a. Physical objects can interact
b. Any device can communicate with any other device
c. Networks integrate seamlessly

Refer Lecture 21@07:34

**QUESTION 4:**

State whether the following statement is true or false

**Statement:** "Use of different programming languages such as JavaScript, Python, JAVA, and others is an example of heterogeneity in IoT. This brings in the need for interoperability."

a. False

b. True

**Correct Answer: b. True**

**Detailed Solution:** Use of different programming languages such as JavaScript, Python, JAVA, and others is an example of heterogeneity in IoT. This brings in the need for interoperability

(Please refer Lecture 21@09:12)

**QUESTION 5:**

State True or False.

**Statement:** "The interoperability between devices and device users in terms of message formats is called Syntactic Interoperability."

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** The interoperability between devices and device users in terms of message formats is called Syntactic Interoperability.

Refer Lecture 21@17:06.

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**QUESTION 6:**

What is the full form of UMB in IoT interoperability?

a. Universal Middleware Bridge

b. Universal Main Bridge

c. Universal Main Bracket

d. None of these

**Correct Answer: a. Universal Middleware Bridge**

**Detailed Solution:** UMB stands for Universal Middleware Bridge.

Refer Lecture 21@22:16.

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**QUESTION 7:**

Which of the following pins is NULL in the DHT-11 sensor?

a. Pin 2

b. Pin 4

c. Both (a) and (b)

d. Neither (a) nor (b)

**Correct Answer: d. Neither (a) nor (b)**

**Detailed Solution:** Pin 3 is NULL in the DHT-11 sensor.

Refer Lecture 24@03:26.

---

**QUESTION 8:**

Arduino MEGA microcontroller board is also known as?

a. MEGA 3370

b. MEGA 2570

c. MEGA 9980

d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** Arduino MEGA microcontroller board is also known as MEGA 2560.

Refer Lecture 24@03:43.

**QUESTION 9:**

A _____ provides high-precision rotary motion of 0 to 180 degrees.

a. Solenoid Valve

b. Relay

c. Servo Motor

d. None of these

**Correct Answer: c. Servo Motor**

**Detailed Solution:** A Servo Motor provides high-precision rotary motion of o to 180 degrees.

Refer Lecture 25@04:47.

**QUESTION 10:**

What does the following code snippet do in interfacing a servo motor with the Arduino MEGA board?

*int servoPin = 12;*

a. Declares pin for connecting servo motor

b. Declares pin for providing power to MEGA board

c. Declares pin for Ground supply for servo motor

d. None of these

**Correct Answer: a. Declares pin for connecting servo motor**

**Detailed Solution:** *int servoPin = 12;* Declares pin for connecting servo motor.

Refer Lecture 25@07:54.

**QUESTION 11:**

What does the *delay(1000);* line do mean in an Arduino sketch?

a. It introduces a delay of 1 second

b. It introduces a delay of 10 seconds

c. It introduces a delay of 100 milliseconds

d. None of these

**Correct Answer: a. It introduces a delay of 1 second**

**Detailed Solution:** The line *delay(1000);* in an Arduino sketch introduces a delay of 1 second.

Refer Lecture 25@08:25

**QUESTION 12:**

State True or False.

**Statement:** "Additional hardware is required to load a program into the Arduino Uno board using a computer."

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** No additional hardware is required to load a program into the Arduino Uno board using a computer.

Refer Lecture 22@05:11

**QUESTION 13:**

Arduino Uno board has _____ number of PWM pins?

a. 9
b. 4
c. 8
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** Arduino Uno board has 6 number of PWM pins.

Refer Lecture 22 @ 07:13.

**QUESTION 14:**

Consider the following Arduino sketch.

```
int r = 4;
int g = 5;
void setup(){

Serial.begin(9600);
pinMode(r,INPUT);
pinMode(g,__??__);
}

void loop()
{
  int val = analogRead(r);
  if( val > 10)
  {
      digitalWrite(g,HIGH);
  }
  else
  {
      digitalWrite(g,HIGH);
  }
  delay(500);
}
```

What must be inserted in the place of '??' within the second pinMode() function in void setup()?

a. INPUT

b. OUTPUT

c. None of these

d. Anyone of these is okay

**Correct Answer: b. OUTPUT**

**Detailed Solution:** Since pin 'g' is writing digital values, it is required to set in OUTPUT mode. Refer Lecture 22-23.

**QUESTION 15:**

State True or False.

**Statement:** "The "Verify" option in the Arduino IDE checks the code for compilation errors."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** The "Verify" option in the Arduino IDE checks the code for compilation errors.

Refer Lecture 24.

**********END**********

# Introduction to Internet of Things

## Assignment-Week 6

**TYPE OF QUESTION:MCQ/MSQ**

**Number of questions:15** **Total marks: 15 X 1= 15**

---

**QUESTION 1:**

State True or False.

**Statement:** "Python is popular for embedded application development as it is a very lightweight programming language."

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** Python is popular for embedded application development as it is a very lightweight programming language.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 1:22)

---

**QUESTION 2:**

State True or False.

**Statement:** "Python does not support strict rules for syntax."

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution:** Python does not support strict rules for syntax.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 03:38)

---

**QUESTION 3:**

_____ is a software that is used to write codes, integrate several modules and libraries.

a. IDE
b. Module
c. API
d. None of these

**Correct Answer: a. IDE**

**Detailed Solution:** IDE is a software that is used to write codes, integrate several modules and libraries.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 05:02).

---

**QUESTION 4:**

State True or False.

**Statement:** "To indicate different blocks of code, Python follows rigid indentation."

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** To indicate different blocks of code, Python follows rigid indentation.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 7::29).

---

**QUESTION 5:**

What is the output of the following line of code in Python?

*>>> print "Hi, Welcome to python!"*

a. Hi, Welcome to python!

b. "Hi, Welcome to python!"

c. Hi, Welcome to python

d. None of these

**Correct Answer: a.** Hi, Welcome to python!

**Detailed Solution:** The output of the following line of code in Python -

*>>> print "Hi, Welcome to python!"*

*Output: Hi, Welcome to python!*

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 07:31)

---

**QUESTION 6:**

Which of the following is a mode to open a file in Python?

a. Read mode
b. Write mode
c. Append mode
d. All of these

**Correct Answer: d.** All of these

**Detailed Solution:** The four basic modes to open a file in python are –

a. Read mode
b. Write mode
c. Append mode
d. Both read and write mode

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 03:00)

---

**QUESTION 7:**

State whether the following command to install the PIL library is correct or not.

*sudo pip get pillow*

a. Correct
b. Incorrect

**Correct Answer: b. Incorrect**

**Detailed Solution:** The command to install the PIL library is *sudo pip install pillow*.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 17:40 )

---

**QUESTION 8:**

How many data types are these in Python?

a. 5

b. 7

c. 8

d. None of these

**Correct Answer: a. 5**

**Detailed Solution:** There are 5 data types in Python.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @9:40).

---

**QUESTION 9:**

State whether the following controlling statement structure is valid in Python or not.

*if(condition):*

*statement 1*

*statement2*

*elif(condition)*

*statement 1*

*statement 2*

*else*

*statement 1*

*statement 2*

a. Invalid
b. Valid

**Correct Answer: a. Invalid**

**Detailed Solution:** Invalid as the colons are missing in elif and else statements.

(Please refer to lecture INTRODUCTION TO RASPBERRY PI-I @29:37).

---

**QUESTION 10:**

Can we configure Raspberry Pi as a Web Server?

a. Yes
b. No

**Correct Answer: a. Yes**

**Detailed Solution:** We can configure Raspberry Pi as a Web Server.

See lecture INTRODUCTION TO RASPBERRY PI-I @ 02:46

---

**QUESTION 11:**

What is the frequency of the Raspberry Pi Zero CPU?

a. 2.1 GHz
b. 1.1 GHz
c. 3.1 Ghz
d. None of these

**Correct Answer: d. None of these**

**Detailed Solution:** The frequency of the Raspberry Pi Zero CPU is 1 GHz.

See lecture INTRODUCTION TO RASPBERRY PI-II @ 05:05

---

**QUESTION 12:**

What does the following line of code do?

*sudo apt-get install python-rpi.gpio*

a. It installs the Raspberry Pi GPIO library
b. It installs the Python essentials for GPIO
c. Both (a) and (b)
d. Neither (a) nor (b)

**Correct Answer: a. It installs the Raspberry Pi GPIO library**

**Detailed Solution:** It installs the Raspberry Pi GPIO library.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-II @ 07:35

---

**QUESTION 13:**

Which command Exits the nano editor?

a. Ctrl + X
b. Ctrl + O
c. Ctrl + K
d. None of these

**Correct Answer: a. Ctrl + X**

**Detailed Solution: Ctrl + O exits the nano editor.**

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-II @ 10:20

---

**QUESTION 14:**

What does the following line of code do?

*GPIO.output(11,True)*

a. Sets GPIO 11 as output pin
b. Turns on GPIO pin 11
c. Neither (a) nor (b)
d. Both (a) and (b)

**Correct Answer: b. Turns on GPIO pin 11**

**Detailed Solution:** Turns on GPIO pin 11

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 11:18)

---

**QUESTION 15:**

What does the following line of code do?

*raspistill -o image.jpg*

a. Captures video feed
b. Captures still image
c. Both (a) and (b)
d. None of these

**Correct Answer: b. Captures still image**

**Detailed Solution:** Captures still image.

(Please refer to lecture INTRODUCTION TO RASPBERYY PI-II @ 19:29)

**********END**********

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**Introduction to**

**Internet of Things**

**Assignment-Week 7**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

**QUESTION 1:**

Adafruit provides a library to work with DHT22 Sensor.

a. True
b. False

**Correct Answer: a. True**
**Detailed Solution:** Adafruit provides a library to work with DHT22 Sensor. (Please refer to lecture Implementation of IoT with Raspberry Pi- II @ 4:41)

**QUESTION 2:**

Which python library is used for plotting the data in 2D.

a. List
b. Numpy
c. Pandas
d. Matplotlib

**Correct Answer: d. Matplotlib**

**Detailed Solution:** Matplotlib is a python library used for plotting the data in 2D. (Please refer to lecture Implementation of IoT with Raspberry Pi- III @ 12:13)

`

**QUESTION 3:**

Fill in the blanks. _____ creates a two-way communication between two nodes in a network.

a. Peer-to-Peer Network

b. Parallel Programming

c. Socket Programming

d. None of these

**Correct Answer: c. Socket Programming**

**Detailed Solution:** Socket Programming creates a two-way communication between two nodes in a network. (Please refer to lecture Implementation of IoT with Raspberry Pi- II @ 6:02).

**QUESTION 4:**

What are the two main challenges of SDN?

a. File placement and Node placement
b. Rule placement and Controller placement
c. Both a and b
d. Neither a nor b

**Correct Answer: b. Rule placement and Controller placement**

**Detailed Solution:** Rule placement and Controller placement are the two main challenges of SDN. (Please refer to lecture Software Defined Networking- Part- I @ 16:49)

`

**QUESTION 5:**

IoT cannot control the network remotely.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** IoT can control the network remotely. (Please refer to lecture Implementation of IoT with Raspberry Pi- II @ 1:53)

**QUESTION 6:**

Which of the following is a part of data processing?

a. Splitting
b. Filtering
c. Neither a nor b
d. Both a and b

**Correct Answer: d. Both a and b**

**Detailed Solution:** Splitting and Filtering are both types of data processing. (Please refer to lecture Implementation of IoT with Raspberry Pi- III @ 7:20)

**QUESTION 7:**

What is introduced if a controller is down in SDN?

a. Backup controller
b. Intro controller
c. Both a and b
d. Neither a nor b

**Correct Answer: a. Backup Controller**

**Detailed Solution:** Backup controller is introduced if a controller is down in SDN. (Please refer to lecture Software Defined Networking- Part- II @ 8:29 )

`

**QUESTION 8:**

Ion() is used to add a subplot in the data plotting in python.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** subplot() is used to add a subplot in the data plotting in python. (Please refer to lecture Implementation of IoT with Raspberry Pi- III @ 23:28).

**QUESTION 9:**

Matplotlib cannot plot in 2D.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** Matplotlib can plot in 2D. (Please refer to lecture Implementation of IoT with Raspberry Pi- III @ 11:54).

**QUESTION 10:**

How many rules are deleted from the switch at hard timeout?

a. None
b. All
c. Half
d. Some

**Correct Answer: b. All**

**Detailed Solution:** All rules are deleted from the switch at hard timeout. See lecture 33 @ 25:02

`

**QUESTION 11:**

Are controllers able to handle all incoming requests from switches?

a. Yes
b. No

**Correct Answer: a. Yes**

**Detailed Solution: Controllers must be able to handle all incoming requests from switches.** See lecture 34 @ 4:05

`

**QUESTION 12:**

Are Indigo and LINC Open source?

a. Yes
b. No

**Correct Answer: a. Yes**

**Detailed Solution: Indigo and LINC are both open source.** See lecture 33@ 26:43.

`

**QUESTION 13:**

Which of the following forward the sensed data based on the ID of the source node?

a. ID-centric data forwarding
b. Northbound IDs
c. Both a and b
d. Neither a nor b

**Correct Answer: a. ID-centric data forwarding**

**Detailed Solution: ID-centric data forwarding forward the sensed data based on the ID of the source node.** See lecture 35 @ 09:49

`

`

**QUESTION 14:**

How many requests can a controller handle through a single thread?

a. 200/sec
b. 50/sec
c. Neither a nor b
d. Both a and b

**Correct Answer: a. 200/sec**

**Detailed Solution: A controller can handle 200 requests in a second through a single thread**. (Please refer to lecture 34@ 3:47)

`

**QUESTION 15:**

Switches that forward the traffic in a distributed manner have a global view of the network.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** Switches that forward the traffic in a distributed manner do not have a global view of the network. (Please refer to lecture 33 @ 7:04)

**********END**********

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# Introduction to Internet of Things

## Assignment-Week 8

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

Which of the following abstracts the physical resources from the network services?

a. Path management
b. Routing protocols
c. Flow-table paradigm
d. Network Virtualization

**Correct Answer: d. Network Virtualization**

**Detailed Solution:** Network Virtualization abstracts the physical resources from the network services (Please refer Lecture 36@3:59)

---

**QUESTION 2:**

In which of these, exact match rules are useful?

a. Elephant Flow

b. Mice-flow

c. Both a and b

d. Neither a nor b

`

**Correct Answer: a. Elephant Flow**

**Detailed Solution: In Elephant Flow, exact match rules are useful.** (Please refer Lecture 36@17:45)

---

**QUESTION 3:**

Which of these takes last k-th location instances to predict the next location?

a. Odin-k Markov Predictor
b. Order-k Markov interpreter
c. Order-k Markov predictor
d. None of these

**Correct Answer: c. Order-k Markov predictor**

**Detailed Solution: Order-k Markov predictor takes last k-th location instances to predict the next location.** (Please refer Lecture 36@15:47)

---

**QUESTION 4:**

Which of the following is an important issue due to the dynamic nature of the IoT network?

a. Match rules
b. Load rules
c. Match balancing
d. Load Balancing

**Correct Answer: d. Load Balancing**

`

**Detailed Solution: Load Balancing is an important issue due to the dynamic nature of the IoT network.** (Please refer Lecture 36 @17:09)

---

**QUESTION 5:**

Fill in the blank.

_____ refers to heterogeneous computing nodes distributed over a wide area to perform very large tasks.

a. Cluster Computing
b. Utility Computing
c. Grid Computing
d. None of these

**Correct Answer: c. Grid Computing**

**Detailed Solution: Grid Computing refers to heterogeneous computing nodes distributed over a wide area to perform very large tasks.** (Please refer Lecture 37 @6:48)

---

**QUESTION 6:**

Which of the following is a type of service model in cloud computing?

a. SaaS
b. PaaS
c. IaaS
d. All of these

**Correct Answer: d. All of these**

`

**Detailed Solution:** All these are a type of service model in cloud computing. (Please refer Lecture 37@25:18)

---

**QUESTION 7:**

Which of the following statements are true about SaaS?

Statement I: Applications are available as 'services'.

Statement II: End-users do not have the control of the cloud infrastructure.

Statement III: Services can be accessed by various types of client devices.

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statement I, II and III

**Correct Answer: d. Statements I, II and III**

**Detailed Solution: All the three statements are correct for the SaaS. Applications are available as 'services'. End-users do not have the control of the cloud infrastructure. Services can be accessed by various types of client devices.**

(Please refer Lecture 37@26:01)

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`

**QUESTION 8:**

Fill in the blank.

_____ means independent of device or location.

a. Scalable
b. Reliability
c. Agile
d. Ubiquitous

**Correct Answer: d. Ubiquitous**

**Detailed Solution: Ubiquitous means independent of device or location.** (Please refer Lecture 37@18:06)

---

**QUESTION 9:**

Fill in the blank.

_____ provides the platform which allows developers to create applications which can be offered as a service via the Internet.

a. Software-as-a-Service
b. Storage-as-a-Service
c. Network-as-a-Service
d. Platform-as-a-Service

**Correct Answer: d. Platform-as-a-Service**

**Detailed Solution: Platform-as-a-Service provides the platform which allows developers to create applications which can be offered as a service via the Internet.** (Please refer Lecture 38@11:35)

---

`

**QUESTION 10:**

Fill in the blank. 'Dynamic scaling of the required infrastructure resources' is a feature of which of the essential characteristic of cloud computing.

a. Scalability

b. Availability

c. Synchronization

d. None of these

**Correct Answer: a. Scalability**

**Detailed Solution: 'Dynamic scaling of the required infrastructure resources' is a feature of scalability characteristics of cloud computing.** (Please refer Lecture 38@7:41)

---

**QUESTION 11:**

Software-as-a-Service does not support a fully pay-as-you-go model.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** Software-as-a-Service supports a fully pay-as-you-go model. (Please refer Lecture 38@13:25)

---

`

**QUESTION 12:**

What is the meaning of the Host Level Security at PaaS and SaaS level?

a. Hide host OS from the end users
b. Security responsibilities are transformed to Cloud service providers
c. Both a and b
d. Neither a nor b

**Correct Answer: c. Both a and b**

**Detailed Solution: Both a and b.** (Please refer Lecture 39@19:23)

---

**QUESTION 13:**

Which of the following is an objective of Data Security?

a. Confidentiality
b. Integrity
c. Availability
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** The three objectives of Data Security are Confidentiality, integrity and availability. (Please refer Lecture 39@22:39)

---

`

**QUESTION 14:**

CloudSim provides a cloud environment.

a. True
b. False

**Correct Answer: a. True**

**Detailed Solution: CloudSim provides a cloud environment.**. (Please refer Lecture 40@8:27)

---

**QUESTION 15:**

Which of the following is true about GreenCloud?

a. GreenCloud cannot monitor the energy consumption of servers, switches, etc.
b. GreenCloud was developed as an extension of GO.
c. Neither a nor b
d. Both a and b

**Correct Answer: c. Neither a nor b**

**Detailed Solution:** GreenCloud monitors the energy consumption of servers, switches, etc. It was developed as an extension of NS2. (Please refer Lecture 40@16:25)

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**********END**********

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**Introduction to Internet of Things Assignment-**

**Week 9**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

**QUESTION 1:**

Which of the following can be used to install OpenStack?

a. devstack
b. stack
c. Swift
d. All of these

**Correct Answer: a. devstack**

**Detailed Solution:** devstack can be used to install OpenStack. (Please refer Lecture 41@8:01)

---

**QUESTION 2:**

Which of the following is/are the advantages of cloud computing?

a. Elasticity
b. Pay-per-use
c. Self Service
d. All of the above

**Correct Answer: d. All of the above**

**Detailed Solution:** The advantages of cloud computing include Elasticity, Pay-per-use and Self-

`

Service. (Please refer Lecture 42@6:45)

---

**QUESTION 3:**

Select the statement(s) that denote the limitations of WSNs.

Statement I: Procurement

Statement II: Deployment

Statement III: Maintenance

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

**Correct Answer: c. Statements I, II, and III**

**Detailed Solution:** The limitations of WSNs are Procurement, Deployment and Maintenance. (Please refer Lecture 42@10:52)

---

`

**QUESTION 4:**

Which of the following plays a role in the sensor-cloud architecture?

a. End-users

b. Sensor-owner

c. Sensor-Cloud Service Provider

d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** End users, sensor-owner, sensor-cloud service provider are the actors in sensor-cloud. (Please refer Lecture 42@14:18)

---

**QUESTION 5:**

The managerial role is played by _____ in sensor-cloud architecture.

a. End-users

b. Sensor-Cloud Service Provider

c. Neither a nor b

d. Both a and b

**Correct Answer: b. Sensor-Cloud Service Provider**

**Detailed Solution: Sensor-Cloud Service Provider plays the managerial role in**

`

**sensor-cloud architecture.** (Please refer Lecture 42@14:29)

---

**QUESTION 6:**

Which of the following is/are the components of the work-flow of the sensor-cloud.

a. Virtual Sensor Manager

b. Virtual Sensor Controller

c. Resource Manager

d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** Virtual Sensor Manager, Virtual Sensor Controller and Resource Manager are all of the components of the work-flow of the sensor-cloud. (Please refer Lecture 42@17:02)

---

**QUESTION 7:**

Who coined the term Fog computing?

a. IBM

b. CISCO

c. Both a and b

d. Neither a nor b

`

**Correct Answer: b. CISCO**

**Detailed Solution: CISCO coined the term Fog computing.** (Please refer Lecture 44@4:18)

---

**QUESTION 8:**

What of the following is responsible for allocation and deallocation of the resources?

a. Sensor manager

b. Sensor controller

c. Sensor resource pool

d. None of these

**Correct Answer: c. Sensor resource pool**

**Detailed Solution:** Sensor resource pool is responsible for allocation and deallocation of the resources (Please refer Lecture 42@16:11)

---

**QUESTION 9:**

The sensor owner can only deploy homogeneous physical sensor nodes in the sensor cloud.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution:** The sensor owner can deploy homogeneous and heterogeneous physical sensor nodes in the sensor cloud. (Please refer Lecture 42@15:03)

`

---

**QUESTION 10:**

What is the function of Internal Cache?

a. Handles requests from end-user
b. Takes decision whether the data should be provided directly to the end user
c. Neither a nor b
d. Both a and b

**Correct Answer: d. Both a and b**

**Detailed Solution:** Internal Cache handles requests from end-user and also takes decision whether the data should be provided directly to the end user (Please refer Lecture 43@21:06)

---

**QUESTION 11:**

What is the other term for fog computing?

a. Foglog

b. Cloud

c. Dew

d. Fogging

**Correct Answer: d. Fogging**

**Detailed Solution:** The other term for fog computing is Fogging. (Please refer Lecture 44@4:19)

---

`

**QUESTION 12:**

Fill in the blanks. Fog computing is an intermediate layer between _____ and _____.

a. Dew and devices

b. Cloud and devices

c. Cloud and server

d. None of these

**Correct Answer: b. Cloud and devices**

**Detailed Solution:** Fog computing is an intermediate layer between Cloud and devices. (Please refer Lecture 44@6:40)

---

**QUESTION 13:**

Bandwidth refers to the bit-rate of data during analysis.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution:** Bandwidth refers to the bit-rate of data during transmission. (Please refer Lecture 44@17:03)

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`

**QUESTION 14:**

Which of the following is used by the fog applications to increase the speed of service accessibility?

a. Low power

b. Good network connection

c. Accelerators

d. Analytics

**Correct Answer: c. Accelerators**

**Detailed Solution:** Accelerators increase the speed of service accessibility. (Please refer to Page 260, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

---

**QUESTION 15:**

Fill in the blank. The concept of _____ enables physical hardware to be shared among multiple entities.

a. Hardware virtualization

b. Software virtualization

c. Module virtualization

d. All of these

**Correct Answer: a. Hardware virtualization**

`

**Detailed Solution:** The concept of hardware virtualization enables physical hardware to be shared among multiple entities. (Please refer to Page 262, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

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**********END**********

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**Introduction to**

**Internet of Things**

**Assignment-Week 10**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

`

**QUESTION 1:**

Smart city is an _____ system.

a. local
b. rural
c. home
d. urban

**Correct Answer: d. urban**

**Detailed Solution:** Smart city is an urban system. (Please refer Lecture 46@8:02)

_____

**QUESTION 2:**

Which of the following is an area of focus in smart parking lots?

a. Auto-routing of vehicles
b. Auto-charging of vehicles
c. Detection of slots in parking lots
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution: Auto-routing of vehicles, auto-charging of vehicles and detection of slots in parking lots**. (Please refer Lecture 46@19:42)

_____

`

**QUESTION 3:**

Select the statement(s) that are true for smart health.

Statement I: Smart health is low-cost and portable.

Statement II: Smart health allows remote check-ups and diagnosis.

Statement III: Smart health is effortless health monitoring.

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

**Correct Answer: c. Statements I, II, and III**

**Detailed Solution:** Smart health is low-cost, portable and effortless health monitoring. Smart health allows remote check-ups and diagnosis. (Please refer Lecture 46@26:12)

---

`

**QUESTION 4:**

What refers to combining information from multiple sensor sources?

a. Information Collection

b. Multi-sensor deployment

c. Multi-sensor dissemination

d. Multi-sensor data fusion

**Correct Answer: d. Multi-sensor data fusion**

**Detailed Solution:** Multi-sensor data fusion combines information from multiple sensor sources. (Please refer Lecture 47@6:52)

---

**QUESTION 5:**

The ensemble of all the decisions occur at the _____.

a. Feature level

b. Decision level

c. Pixel level

d. All of these

`

**Correct Answer: b. Decision level**

**Detailed Solution: The ensemble of all the decisions occur at the Decision level.** (Please refer Lecture 47@11:10)

---

**QUESTION 6:**

The smart parking reduces traffic _____.

a. mobility

b. area

c. analysis

d. congestion

**Correct Answer: d. congestion**

**Detailed Solution:** The smart parking reduces traffic congestion. (Please refer Lecture 47@16:58)

---

`

**QUESTION 7:**

Which of the following is the correct full form of DLNA Standards for HAN?

a. Digital Lifestyle Network Alliance

b. Digital Living Network Alliance

c. Digital Lifestyle Network Algorithm

d. None of these

**Correct Answer: b. Digital Living Network Alliance**

**Detailed Solution: DLNA stands for Digital Living Network Alliance standards.** (Please refer Lecture 48@13:49)

---

**QUESTION 8:**

Which of the following is a type of sensing?

a. Stationary

b. Mobile

c. Both a and b

d. Neither a nor b

`

**Correct Answer: c. Both a and b**

**Detailed Solution:** Stationary and Mobile are two types of sensing. (Please refer Lecture 47@19:36)

---

**QUESTION 9:**

Smart home infrastructure cannot include wireless devices.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution:** Smart home infrastructure can include wireless/wired devices. (Please refer Lecture 48@6:17)

---

**QUESTION 10:**

Which of the following is a layer of Zigbee ?

a. Physical layer
b. MAC Layer
c. Network Layer
d. All of these

`

**Correct Answer: d. All of these**

**Detailed Solution:** The layers of Zigbee are Physical layer, MAC Layer, Network Layer are all the layers of Zigbee. (Please refer Lecture 48@16:23)

---

**QUESTION 11:**

Which of the following is a phase of ICV development?

a. Based on 2G

b. Based on 4G LTE

c. Vehicles connected to cloud

d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** The phases of ICV development: Phase 1: Based on 2G, Phase 2: Based on 4G LTE, Phase 3: Vehicles connected to cloud (Please refer Lecture 50@12:28)

---

`

**QUESTION 12:**

Which of the following is a property of V2X?

a. Restricted to a single source

b. Distributed architecture

c. Both of a and b

d. Neither a nor b

**Correct Answer: b. Distributed architecture**

**Detailed Solution:** V2X follows a distributed architecture. (Please refer Lecture 49@12:06)

---

**QUESTION 13:**

The vehicles cannot be mobile in the Ad-hoc domain of VANET.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution:** The vehicles are mobile in the Ad-hoc domain of

`

VANET. (Please refer Lecture 50@16:59)

---

**QUESTION 14:**

Which of the following is a domain of VANET?

a. In-vehicle

b. Ad-hoc

c. Both a and b

d. Neither a nor b

**Correct Answer: c. Both a and b**

**Detailed Solution:** Both of the In-vehicle and Ad-hoc are the two domains of VANETs. (Please refer Lecture 50@14:36)

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`

**QUESTION 15:**

Fill in the blank. ICV is based on _____.

a. Dedicated Short Range Communication

b. Directed Short Range Communication

c. Dedicated Small Range Communication

d. All of these

**Correct Answer: a. Dedicated Short Range Communication**

**Detailed Solution:** ICV is based on Dedicated Short Range Communication.

(Please refer Lecture 50@7:22)

_____

**********END**********

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**Introduction to**

**Internet of Things**

**Assignment-Week 11**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

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**QUESTION 1:**

What are some of the other names of the smart grids?

a. Energy internet
b. Electricity with a hand
c. Electrolyte
d. All of these

**Correct Answer: a. Energy internet**

**Detailed Solution:** Smart grids are also known as Electricity with a brain, Energy internet, and Electronet. (Please refer Lecture 51@8:33)

_____

`

**QUESTION 2:**

Smart Grid follows which type of flow of energy?

a. Bidirectional

b. Unidirectional

c. Both a and b

d. Neither a nor b

**Correct Answer: a. Bidirectional**

**Detailed Solution: Smart grid follows bidirectional flow of energy.** (Please refer Lecture 51@5:41)

---

**QUESTION 3:**

Which of the following is a benefit associated with smart grids for customers?

a. Reduced transmission of electricity
b. Higher electricity rates
c. Less security
d. Different pricing options

**Correct Answer: d. Different pricing options**

`

**Detailed Solution: Smart Grids provide different pricing options for customers, lower electricity rates and Improved security.** (Please refer Lecture 51@16:33)

---

**QUESTION 4:**

Which of the following is a property of smart grid related to power system efficiency?

a. Power monitoring
b. Asset Management
c. Distribution automation
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution: Power monitoring, Asset Management and Distribution automation are all the properties of smart grid related to power system efficiency.** (Please refer Lecture 51 @22:21)

---

**QUESTION 5:**

Fill in the blank.

In _____, a home can have power from distributed resources.

a. Peaking
b. Islanding
c. Neither a nor b
d. Both a and b

`

**Correct Answer: b. Islanding**

**Detailed Solution:** In islanding, a home can have power from distributed resources.(Please refer Lecture 51 @31:20)

---

**QUESTION 6:**

Which of the following aggregate the energy consumption or energy request of a certain geographical area?

a. DAUs
b. Meter Data Management System
c. Buffer
d. None of these

**Correct Answer: a. DAUs**

**Detailed Solution: DAUs** aggregate the energy consumption or energy request of a certain geographical area. (Please refer Lecture 52@15:17)

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**QUESTION 7:**

Which of the following statements are true about Dynamic System Attacks?

Statement I: Replay attacks compromises sensors and monitors their outputs.

Statement II: D-DIA can lead to a system collapse.

Statement III: Covert attack is a closed loop version of replay attacks.

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statement I, II and III

**Correct Answer: d. Statements I, II and III**

**Detailed Solution:** Replay attacks compromises sensors and monitors their outputs. D-DIA can lead to a system collapse. Covert attack is a closed loop version of replay attacks. (Please refer Lecture 52@20:03)

---

**QUESTION 8:**

Fill in the blank.

_____ is performed by compromising the availability of the grid component.

a. Gateway
b. Data injection attacks
c. AMH
d. Denial of service

**Correct Answer: d. Denial of service**

`

**Detailed Solution: Denial of service** is performed by compromising the availability of the grid component. (Please refer Lecture 52@19:52)

---

**QUESTION 9:**

Fill in the blank.

IIoT is a network of _____ .

a. Physical objects
b. Systems
c. Platforms
d. All of these

**Correct Answer: d. All of these**

**Detailed Solution: IIoT is a network of Physical objects, Systems and Platforms.** (Please refer Lecture 52@14:46)

---

**QUESTION 10:**

Fill in the blank. _____ is based on a wrap and re-use approach.

a. AoT

b. PoT

c. ToII

d. IIoT

**Correct Answer: d. IIoT**

`

**Detailed Solution: IIoT is based on the wrap and re-use approach.** (Please refer Lecture 53@9:51)

---

**QUESTION 11:**

Mechanized production is a part of the 4th Industrial Revolution.

a. True
b. False

**Correct Answer: b. False**

**Detailed Solution:** Mechanized production is a part of the 1st Industrial Revolution. (Please refer Lecture 53@10:41)

---

**QUESTION 12:**

Which of the following happened in the 3rd Revolution?

a. Mass production
b. IIoT
c. Mechanized production
d. None of these

`

**Correct Answer: d. None of these**

**Detailed Solution: Internet evolution and automation happened in the 3rd Revolution.**(Please refer Lecture 53@10:06)

---

**QUESTION 13:**

Which of the following is used to improve the interoperability of different systems?

a. Standardization
b. Optimization
c. Monitoring
d. None of these

**Correct Answer: a. Standardization**

**Detailed Solution: Standardization** improves the interoperability of different systems. (Please refer Lecture 54@16:09)

---

**QUESTION 14:**

Big data is characterized by 7 Vs.

a. True
b. False

**Correct Answer: a. True**

`

**Detailed Solution:**Big data is characterized by 7 Vs. (Please refer Lecture 55@10:36)

---

**QUESTION 15:**

What is Hadoop used in?

a. Distributed processing of large datasets
b. Small clusters of computers
c. Both a and b
d. Neither a nor b

**Correct Answer: a. Distributed processing of large datasets**

**Detailed Solution: Hadoop is used for** distributed processing of large datasets across Large clusters of computers (Please refer Lecture 55@23:18)

---

**********END**********

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# Introduction to Internet of Things

## Assignment-Week 12

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions**: **15** **Total marks: 15 X 1= 15**

**QUESTION 1:**

Qualitative analysis does not deal with the analysis of data that is categorical in nature.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution: Qualitative analysis deals with the analysis of data that is categorical in nature.** (Please refer Lecture 56@2:43)

---

**QUESTION 2:**

Fill in the blank.

Examples of dispersion measures include _____ and _____.

a. Discrete, categorical
b. Continuous, quantitative
c. Discrete, quantitative
d. Range, Variance

**Correct Answer: d. Range, Variance**

**Detailed Solution:** Examples of dispersion measures include Range and Variance (Please refer Lecture 56@13:51)

---

**QUESTION 3:**

Select the statement(s) that denote the key variables of determining statistical significance.

Statement I: Range size

Statement II: Effect size

Statement III: Sample size

a. Statement I

b. Statement I and II

c. Statements II, and III

d. None of these

**Correct Answer: c. Statements II, and III**

**Detailed Solution:** The key variables of determining statistical significance include effect size and sample size (Please refer Lecture 56@16:32)

---

**QUESTION 4:**

The different objectives of AgriSens includes –

a. More yields with less water

b. Save limited water resource

c. Automatic irrigation

d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** The different objectives of AgriSens includes More yields with less water, Save limited water resources and Automatic irrigation. (Please refer Lecture 57@5:32)

---

**QUESTION 5:**

The two most relevant sensors directly used in agriculture are _____.

a. Soil moisture and ECG sensor

b. Soil moisture and water level sensor

c. ECG sensor and water level sensor

d. All of these

**Correct Answer: b. Soil moisture and water level sensor**

**Detailed Solution:** Soil moisture and water level sensors are the necessary sensors generally used in agriculture. (Please refer Lecture 57@13:11)

---

**QUESTION 6:**

The integrated design of the remote server includes_____.

a. Data server

b. Web server

c. Multi users server

d. All of these

**Correct Answer: d. All of these**

**Detailed Solution:** The integrated design of the remote server includes repository data server, web server and multi-users server. (Please refer Lecture 57@9:35)

**QUESTION 7:**

Select the correct order of the component layers present in the IoT healthcare.

a. Sensing layer, cloud platform layer, aggregated layer, processing layer

b. Sensing layer, aggregated layer, processing layer, cloud platform layer

c. Aggregated layer, sensing layer, processing layer, cloud platform layer

d. Sensing layer, processing layer, aggregated layer, cloud platform layer

**Correct Answer: b. Sensing layer, aggregated layer, processing layer, cloud platform layer**

**Detailed Solution:** The sensing layer senses data and transmit it o the aggregation layer where the data are aggregated. The aggregated layer further transfers the data to the processing layer the data are processed and final sent to the cloud platform. (Please refer Lecture 58@8:25)

---

**QUESTION 8:**

Which of the following networks is used in AmbuSens?

a. IEEE 802.15.4

b. 6LoWPAN

c. Wireless Body Area Network

d. IEEE 802.15.2

**Correct Answer: c. Wireless Body Area Network**

**Detailed Solution:** In AmbuSens, the network used is Wireless Body Area Network. (Please refer Lecture 58@23:03)

---

**QUESTION 9:**

Wireless IoT driven solutions for remote healthcare facility provisioning brings healthcare to patients than bringing patients to healthcare.

a. True

b. False

**Correct Answer: a. True**

**Detailed Solution:** In IoT health care, wireless IoT driven solutions bring healthcare to patients than bringing patients to healthcare. (Please refer Lecture 58@11:40)

---

**QUESTION 10:**

Fill in the blank. The statistical data analysis tools can be applied on_____.

a. Sensor data

b. Images

c. Videos

d. All of these

**Correct Answer: a. Sensor data**

**Detailed Solution:** The statistical data analysis tools can be applied on sensor data. Please refer Lecture 59@18:16)

---

**QUESTION 11:**

Which of the following sensors are present in smartphones handheld devices are used for activity monitoring?

a. EEG and GPS

b. Accelerometer and EEG

c. Accelerometer and Gyroscope

d. All of these

**Correct Answer: c. Accelerometer and Gyroscope**

**Detailed Solution: Accelerometer and Gyroscope are used for activity monitoring.** (Please refer Lecture 59@15:29)

---

**QUESTION 12:**

Which of the following is the primary disadvantage of using camera-based activity tracking?

a. Accurate

b. Processing intensive

c. Both Accurate and Processing intensive

d. None of these

**Correct Answer: b. Processing intensive**

**Detailed Solution:** Camera based activity tracking is very much process intensive. (Please refer Lecture 59@17:43)

---

**QUESTION 13:**

By performing continuous monitoring of a person's activity, it is not possible to observe his/her behavior or to identify any repetitive pattern in his/her day-to-day activity.

a. True

b. False

**Correct Answer: b. False**

**Detailed Solution:** Continuous monitoring of activity results in daily observations of human behavior and repetitive patterns in their activity. (Please refer Lecture 59@14:33)

---

**QUESTION 14:**

Which of the following are the inbuilt sensors that are present in high end smartphones?

a. ECG and EEG

b. Accelerometer, proximity sensor, and EEG

c. Accelerometer, Compass, and Gyroscope

d. Pressure sensor and NPK sensor

**Correct Answer: c. Accelerometer, Compass, and Gyroscope**

**Detailed Solution:** High end smartphones are likely to have accelerometer, compass, and gyroscope. (Please refer Lecture 60@2:28)

---

**QUESTION 15:**

Fill in the blank. Processing the handheld activity device data with artificial intelligence can be used for _____.

a. Fall detection

b. Heart rate detection

c. Vehicle detection

d. All of these

**Correct Answer: a. Fall detection**

**Detailed Solution:** Processing the handheld activity device data with artificial intelligence can be used for detecting sudden fall of a person. (Please refer Lecture 60@11:56)

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**********END**********